IMPLICATIONS AMONG THE NINE BOUNDEDNESS AND POINT CONDITIONS ON ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. The four diamond edges are definition-containment checks carried as explicit obligations; the zeta-to-epsilon edge is a cross-chapter citation; the base-change input is not carded ('clear from the definition') and is declared external.

Allowed standard prerequisites (editorial).

- Very-reasonable characterization. lemma-characterize-very-reasonable (carded L26702). Expands (epsilon) into the Zariski-local covering family.
- Descent of universal boundedness. lemma-descent-universally-bounded (carded L26696). Lifts boundedness from W -> X_i to W -> X in the epsilon case.
- Base change of universal boundedness. lemma-base-change-universally-bounded (same chapter, NOT carded, declared external). Gives boundedness of the projections R -> U in the delta case.
- Composition and finiteness of bounded fibres. lemma-composition-universally-bounded (same chapter, carded L26049) and lemma-universally-bounded-finite-fibres (same chapter, carded L26058). Complete the delta-to-gamma chain.
- Quasi-separated spaces are very reasonable. Properties of Spaces, lemma quasi-separated-quasi-compact-pieces (cross-chapter, named). Gives (zeta) implies (epsilon).

Source reference. Original lemma 03JU: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 534–534. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JS: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 485–485. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JT: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 847–847. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JV: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 732–732. Navigation pointer only: the counted proof cites

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this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JN: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 327–327. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03IH: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 896–896. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JM: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 297–297. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JO: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 342–342. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JQ: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 417–417. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03JA: focused-seeds/geometry/S9-decent-bounded-fibres/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

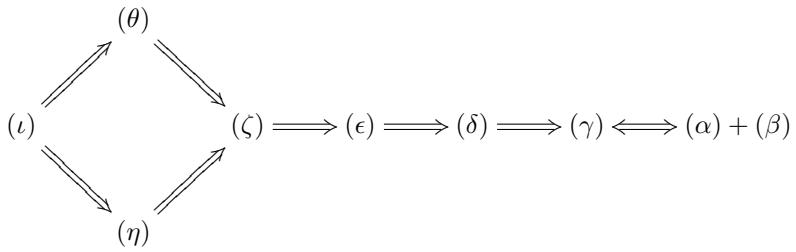
Source reference. Original reference 03W7: focused-seeds/geometry/S9-decent-bounded-fibres/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Consider the following conditions on X:*

(α) *For every $x \in |X|$, the equivalent conditions of Lemma 03JS hold.*
(β) *For every $x \in |X|$, the equivalent conditions of Lemma 03JU hold.*
(γ) *For every $x \in |X|$, the equivalent conditions of Lemma 03JV hold.*
(δ) *The equivalent conditions of Lemma 03JT hold.*
(ϵ) *The equivalent conditions of Lemma 03IH hold.*

(ζ) *The space X is Zariski locally quasi-separated.*
(η) *The space X is quasi-separated*
(θ) *The space X is representable, i.e., X is a scheme.*
(ι) *The space X is a quasi-separated scheme.*

We have



Proof. The implication $(\gamma) \Leftrightarrow (\alpha) + (\beta)$ is immediate. The implications in the diamond on the left are clear from the definitions.

Assume (ζ), i.e., that X is Zariski locally quasi-separated. Then (ϵ) holds by Properties of Spaces, Lemma 03W7.

Assume (ϵ). By Lemma 03IH there exists a Zariski open covering $X = \bigcup X_i$ such that for each i there exists a scheme U_i and a quasi-compact surjective étale morphism $U_i \to X_i$. Choose an i and an affine open subscheme $W \subset U_i$. It suffices to show that $W \to X$ has universally bounded fibres, since then the family of all these morphisms $W \to X$ covers X. To do this we consider the diagram

$$\begin{array}{ccc} W \times_X U_i & \xrightarrow{p} & U_i \\ {\scriptstyle q}\downarrow & & \downarrow \\ W & \longrightarrow & X \end{array}$$

Since $W \to X$ factors through X_i we see that $W \times_X U_i = W \times_{X_i} U_i$, and hence q is quasi-compact. Since W is affine this implies that the scheme $W \times_X U_i$ is quasi-compact. Thus we may apply Morphisms, Lemma 03JA and we conclude that p has universally bounded fibres. From Lemma 03JO we conclude that $W \to X$ has universally bounded fibres as well.

Assume (δ). Let U be an affine scheme, and let $U \to X$ be an étale morphism. By assumption the fibres of the morphism $U \to X$ are universally bounded. Thus also the fibres of both projections $R = U \times_X U \to U$ are universally bounded, see Lemma 03JN. And by Lemma 03JM also the fibres of $R \to X$ are universally bounded. Hence for any $x \in X$ the fibres of $|U| \to |X|$ and $|R| \to |X|$ over x are finite, see Lemma 03JQ. In other words, the equivalent conditions of Lemma 03JV hold. This proves that $(\delta) \Rightarrow (\gamma)$. □